

5.4 Examples Using the for Statement (cont.)



- ▶ C++ does *not* include an exponentiation operator, so we use the **standard library function pow**.
 - `pow(x, y)` calculates the value of `x` raised to the `yth` power.
 - Takes two arguments of type `double` and returns a `double` value.
- ▶ This program will not compile without including header file `<cmath>`.
 - Includes information that tells the compiler to convert the value of `year` to a temporary `double` representation before calling the function.
 - Contained in `pow`'s function prototype.

5.4 Examples Using the for Statement (cont.)



- ▶ Example:
- ▶ $X^2 \rightarrow \text{pow}(X, 2)$
- ▶ $(Y+1)^4 \rightarrow \text{pow}(Y+1, 4)$

5.4 Examples Using the for Statement (cont.)



► The parallel loop

In for loop you can use more than one control variable. Optionally, using the comma operator (,) we can specify more than one expression in any of the fields included in a for loop, like in initialization, for example. The comma operator (,) is an expression separator, it serves to separate more than one expression where only one is generally expected. For example, suppose that we wanted to initialize more than one variable in our loop:



5.4 Examples Using the for Statement (cont.)

```
for ( n=0, i=100 ; n!=i ; n++, i-- )  
{  
// whatever here...  
}
```

- ▶ This loop will execute for 50 times if neither n or i are modified within the loop
- ▶ n starts with a value of 0, and i with 100, the condition is $n \neq i$ (that n is not equal to i). Because n is increased by one and i decreased by one, the loop's condition will become false after the 50th loop, when both n and i will be equal to 50.

5.4 Examples Using the for Statement (cont.)



► Nested for loops

As usual, it can be either a single statement or a block enclosed in braces { }.

```
for ( initialization expression1 ; condition expression1 ; increment1 )  
    for ( initialization expression2 ; condition expression2 ; increment2 )  
        statement ;
```

this form used for single statement

5.4 Examples Using the for Statement (cont.)



```
#include <iostream>  
using namespace std;
```

```
void main ( )  
{ for (int m=0; m<2; m++)  
    for (int n=10; n>0; n--)  
        cout << n << ", ";  
}
```

the o/p of this program will be:

10, 9, 8, 7, 6, 5, 4, 3, 2, 1, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1

5.4 Examples Using the for Statement (cont.)



- ▶ Test: how to make the output as follows:

10, 9, 8, 7, 6, 5, 4, 3, 2, 1

10, 9, 8, 7, 6, 5, 4, 3, 2, 1

5.4 Examples Using the for Statement (cont.)



```
#include <iostream>  
using namespace std;
```

```
int main ( )  
{ for (int m=0; m<2; m++)  
  for (int n=10; n>0; n--)  
    cout << n << "\n ";  
}
```

- ▶ What Is the output?

5.4 Examples Using the for Statement (cont.)



```
#include <iostream>  
using namespace std;
```

```
void main ( )  
{ for (int m=0; m<2; m++)  
    for (int n=10; n>0; n-=2)  
        cout << n << "|n ";  
}
```

- ▶ What Is the output?



5.5 do...while Repetition Statement

- ▶ Similar to the `while` statement.
- ▶ The `do...while` statement tests the loop-continuation condition *after* the loop body executes; therefore, *the loop body always executes at least once*.
- ▶ Figure 5.7 uses a `do...while` statement to print the numbers 1–10.



```
1 // Fig. 5.7: fig05_07.cpp
2 // do...while repetition statement.
3 #include <iostream>
4 using namespace std;
5
6 int main()
7 {
8     unsigned int counter = 1; // initialize counter
9
10    do
11    {
12        cout << counter << " "; // display counter
13        ++counter; // increment counter
14    } while ( counter <= 10 ); // end do...while
15
16    cout << endl; // output a newline
17 } // end main
```

1 2 3 4 5 6 7 8 9 10

Fig. 5.7 | do...while repetition statement.



5.5 do...while Repetition Statement (cont.)

do...while Statement UML Activity Diagram

- ▶ Figure 5.8 contains the do...while statement's UML activity diagram, which makes it clear that the loop-continuation condition is not evaluated until after the loop performs its body at least once.

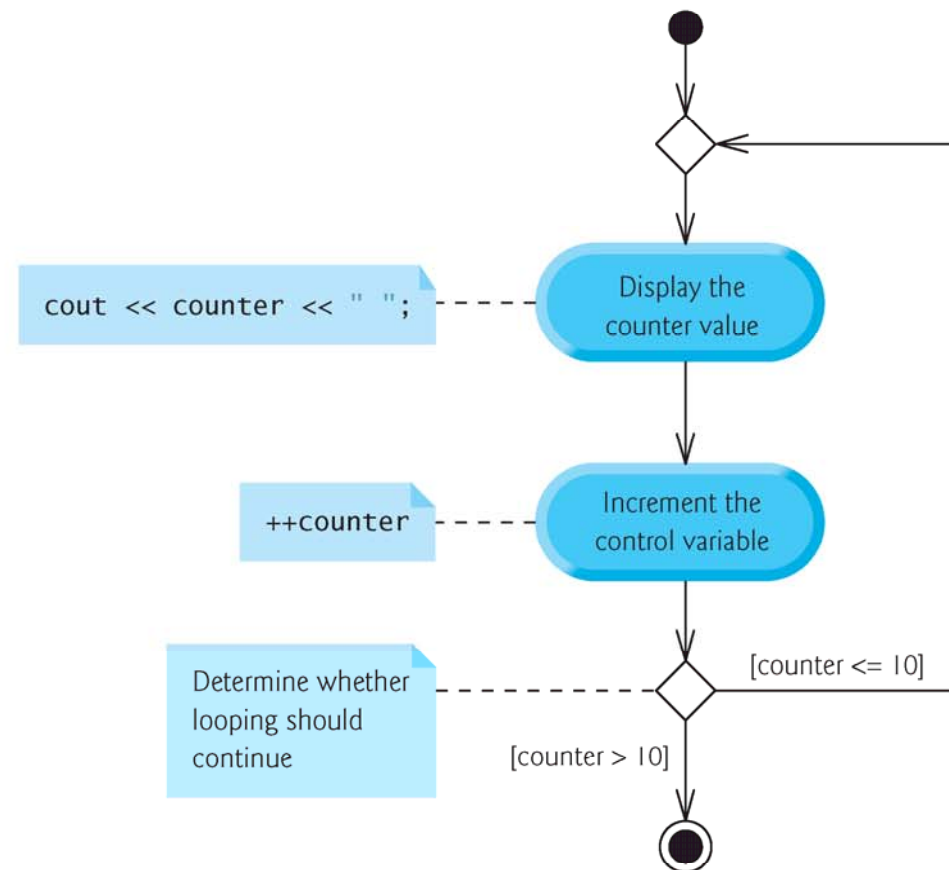


Fig. 5.8 | UML activity diagram for the do...while repetition statement of Fig. 5.7.



5.5 do...while Repetition Statement (cont.)

Braces in a do...while Statement

- ▶ It's not necessary to use braces in the do...while statement if there's only one statement in the body; however, most programmers include the braces to avoid confusion between the `while` and `do...while` statements.
- ▶ For example,
`while ( condition )`
- ▶ normally is regarded as the header of a while statement.



5.5 do...while Repetition Statement (cont.)

- ▶ A `do...while` with no braces around the single statement body appears as

```
do
    statement
while ( condition );
```
- ▶ which can be confusing.
- ▶ You might misinterpret the last line—`while( condition );`—as a while statement containing as its body an empty statement.



5.5 do...while Repetition Statement (cont.)

- ▶ Thus, the `do...while` with one statement often is written as follows to avoid confusion:

```
do
{
    statement
} while ( condition );
```



```
#include <iostream>
using namespace std;
```

```
void main ()
{
```

```
    unsigned long n;
```

```
    do {
```

```
        cout << "Enter number (0 to end):\n ";
```

```
        cin >> n;
```

```
        cout << "You entered: " << n << "\n\n";
```

```
    } while (n != 0);
```

```
}
```



the o/p of this program will be:

Enter number (0 to end): 12345

You entered: 12345

Enter number (0 to end): 160277

You entered: 160277

Enter number (0 to end): 0

You entered: 0



Example comparison between (while, do..while, for)

Example

Write a program to calculate and print the average of a set of integer numbers (N).
Use while and do-while loops:



Example comparison between (while, do..while, for) (cont.)

// using while

```
#include <iostream>
using namespace std;
```

```
void main ()
{
```

```
int x, n, i=1, sum=0;
float avr;
cout<<"n = ";
cin>>n;
```

```
while (i<=n) {
    cout<<"x"<<i<<" = ";
    cin>>x;
    sum += x;
    i++;
}
avg = sum/n;
cout<<"average = "<< avg ;

}
```



Example comparison between (while, do..while, for) (cont.)

// using do-while

```
#include <iostream>
using namespace std;
```

```
void main ()
{
```

```
int x, n, i=1, sum=0;
float avr;
cout<<"n = ";
cin>>n;
```

```
do{
cout<<"x"<<i<<" = ";
cin>>x;
sum += x;
i++;
} while (i<=n);
avr = sum/n;
cout<<"average = "<< avr ;

}
```



Example comparison between (while, do..while, for) (cont.)

//using for loop

```
#include <iostream>
using namespace std;
```

```
void main()
{
```

```
int n, sum = 0, x;
cout<<"n = ";
cin>>n;
```

```
for(int i=1; i<=n; i++)
{
    cout<<"x"<<i<<" = ";
    cin>>x;
    sum += x;
}
```

```
    avg = sum/n;
    cout<<"average = "<<
```

```
avg ;
```

```
}
```